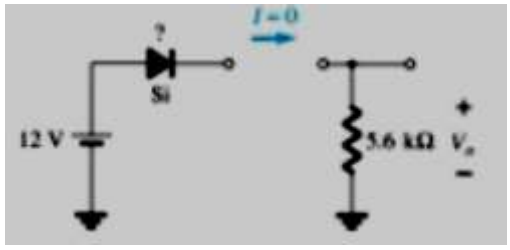
Solutions Tutorial 6

1.



I_D=0

$$V_o = I_R R = I_D R = (0\text{ A})R = 0\text{ V}$$

and $$V_{D_2} = V_{\text{open circuit}} = E = \mathbf{12\text{ V}}$$

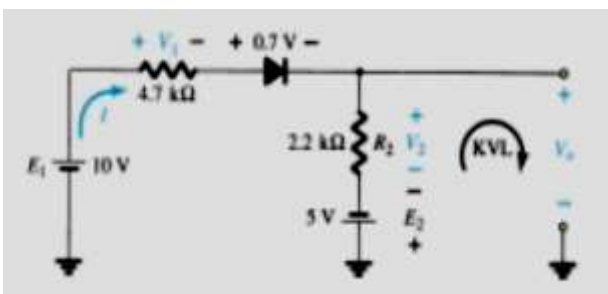
Applying Kirchhoff's voltage law in a clockwise direction gives us

$$E - V_{D_1} - V_{D_2} - V_o = 0$$

and $$V_{D_2} = E - V_{D_1} - V_o = 12\text{ V} - 0 - 0$$
$$= \mathbf{12\text{ V}}$$

with $$V_o = \mathbf{0\text{ V}}$$

2.



$$I = \frac{E_1 + E_2 - V_D}{R_1 + R_2} = \frac{10\text{ V} + 5\text{ V} - 0.7\text{ V}}{4.7\text{ k}\Omega + 2.2\text{ k}\Omega} = \frac{14.3\text{ V}}{6.9\text{ k}\Omega}$$
$$\cong \mathbf{2.072\text{ mA}}$$

and the voltages are

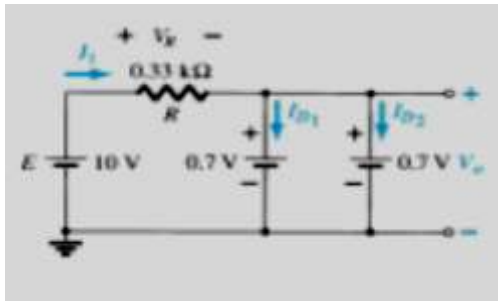
$$V_1 = IR_1 = (2.072\text{ mA})(4.7\text{ k}\Omega) = \mathbf{9.74\text{ V}}$$
$$V_2 = IR_2 = (2.072\text{ mA})(2.2\text{ k}\Omega) = \mathbf{4.56\text{ V}}$$

Applying Kirchhoff's voltage law to the output section in the clockwise direction will result in

$$-E_2 + V_2 - V_o = 0$$

and $$V_o = V_2 - E_2 = 4.56\text{ V} - 5\text{ V} = \mathbf{-0.44\text{ V}}$$

3. V_o=0.7V

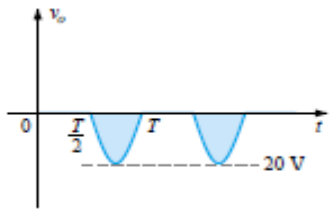


$$I_1 = \frac{V_R}{R} = \frac{E - V_D}{R} = \frac{10\text{ V} - 0.7\text{ V}}{0.33\text{ k}\Omega} = \mathbf{28.18\text{ mA}}$$

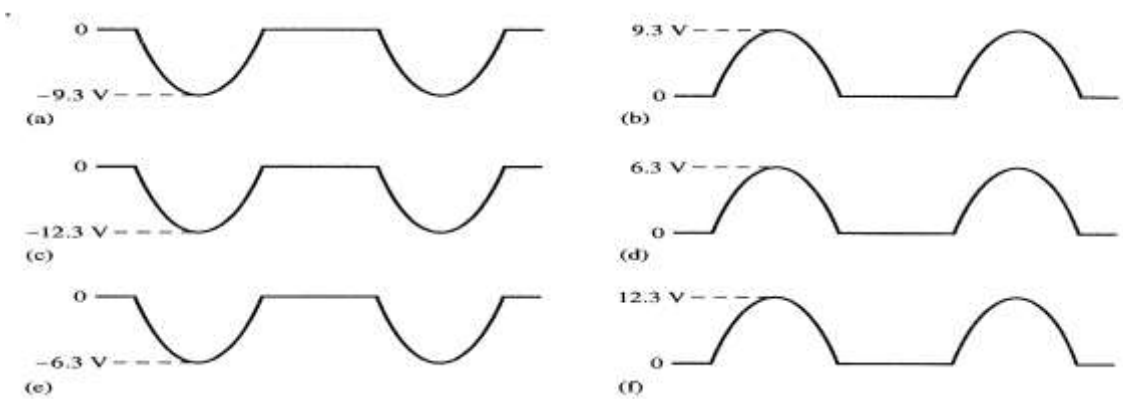
$$I_{D_1} = I_{D_2} = \frac{I_1}{2} = \frac{28.18\text{ mA}}{2} = \mathbf{14.09\text{ mA}}$$

4.

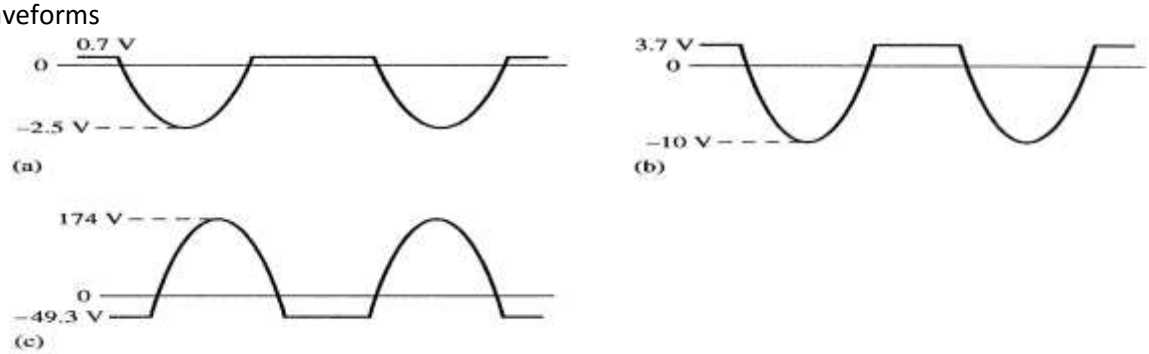
$$V_{dc} = -0.318V_m = -0.318(20\text{ V}) = \mathbf{-6.36\text{ V}}$$



5. Waveforms



6. Waveforms



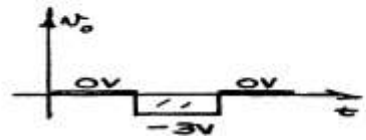
7.

(a) A sine wave with a positive peak at 0.7 V, a negative peak at −7.3 V, and a dc value of −3.3 V.
(b) A sine wave with a positive peak at 29.3 V, a negative peak at −0.7 V, and a dc value of +14.3 V.
(c) A square wave varying from +0.7 V to −15.3 V with a dc value of −7.3 V.
(d) A square wave varying from +1.3 V to −0.7 V with a dc value of +0.3 V.

8.

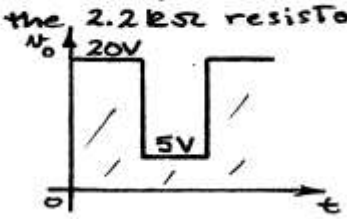
(a)
For $v_i = 20V$ the diode is reverse-biased and $v_o = \underline{0V}$.
For $v_i = -5V$, v_i overpowers the 2V battery and the diode is "on".
Applying Kirchhoff's voltage law in the clockwise direction:

$$-5V + 2V - v_o = 0$$
$$v_o = \underline{-3V}$$



For $v_i = 20V$ the 20V level overpowers the 5V supply and the diode is "on". Using the short-circuit equivalent for the diode we find $v_o = v_i = \underline{20V}$.

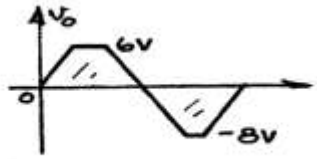
For $v_i = -5V$, both v_i and the 5V supply reverse-bias the diode and separate v_i from v_o. However, v_o is connected directly through the 2.2kΩ resistor to the 5V supply and $v_o = \underline{5V}$



9.

For the positive region of v_i:
The right Si diode is reverse-biased.
The left Si diode is "on" for levels of v_i greater than $5.3V + 0.7V = 6V$. In fact, $v_o = 6V$ for $v_i \geq 6V$
For $v_i < 6V$ both diodes are reverse-biased and $v_o = v_i$.

For the negative region of v_i:
The left Si diode is reverse-biased.
The right Si diode is "on" for levels of v_i more negative than $7.3V + 0.7V = 8V$. In fact, $v_o = -8V$ for $v_i \leq -8V$
For $v_i > -8V$ both diodes are reverse-biased and $v_o = v_i$.



i_R: For $-8V < v_i < 6V$ there is no conduction through the 10 kΩ resistor due to the lack of a complete circuit. Therefore, $i_R = 0mA$

For $v_i \geq 6V$

$v_R = v_i - v_o = v_i - 6V$

For $v_i = 10V$, $v_R = 10V - 6V = 4V$

and $i_R = \frac{4V}{10k\Omega} = 0.4mA$

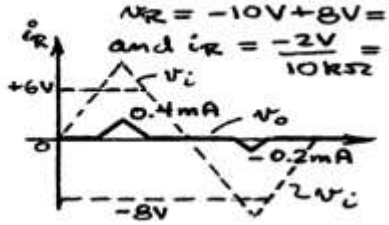
For $v_i \leq -8V$

$v_R = v_i - v_o = v_i + 8V$

For $v_i = -10V$

$v_R = -10V + 8V = -2V$

and $i_R = \frac{-2V}{10k\Omega} = -0.2mA$



10.

(a) $\tau = RC = (56k\Omega)(0.1\mu F) = 5.6ms$

$5\tau = \underline{28ms}$

(b) $5\tau = 28ms >> \frac{T}{2} = \frac{1ms}{2} = \underline{0.5ms}$, 56 : 1

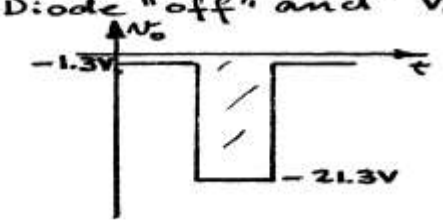
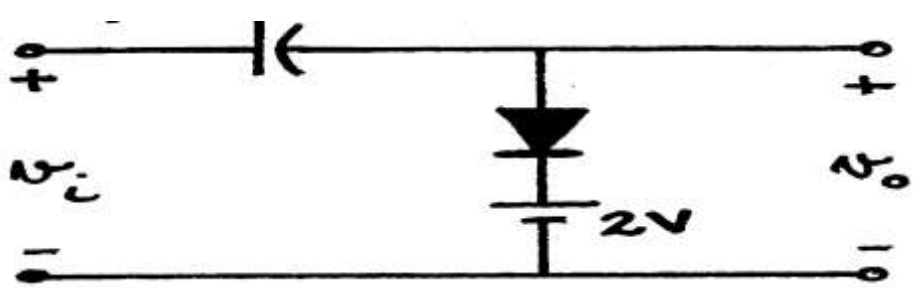
(c) Positive pulse of v_i:

Diode "on" and $V_o = -2V + 0.7V = -1.3V$

Capacitor changes to $10V + 2V - 0.7V = 11.3V$

Negative pulse of v_i:

Diode "off" and $V_o = -10V - 11.3V = -21.3V$



11.